

MAINTENANCE MANUAL

Machine Relocation Detector

Battery Replacement Procedure Manual

Manual No.: HGENMA0114E
NC: MAZATROL Smooth series
MAZATROL MATRIX series
MAZATROL MATRIX NEXUS series
MAZATROL SMART
MAZATROL FUSION 640TX

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. This manual was considered complete and accurate at the time of publication, however, due to our desire to constantly improve the quality and specification of all our products, it is subject to change or modification. If you have any questions, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.

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SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

1 INTRODUCTION

This manual explains what to do if the machine relocation detector raises an alarm, and also important things you should keep in mind when replacing the batteries of the machine relocation detector. If the alarm indicating drop in voltage of the battery is displayed, follow the instructions given in this manual and replace the battery with new one specified by Mazak as soon as possible. Otherwise, the automatic operation of the machine becomes impossible.

Do not start replacing the batteries until you fully understand the contents of this manual.

1-1 Background of the Installation of Machine Relocation Detectors

The machine relocation detector installed in the machine is designed to detect the replacement of the machine and to restrict its operation if it is replaced.

The background of the installation of machine relocation detectors is that certain machines were illegally exported by some Japanese manufacturers to third country as the proliferation of nuclear weapons, weapons of mass destruction, and so on has been a global issue recently. In view of this, the Japanese government requested each Japanese machine tool manufacturer to study the feasibility of installing machine relocation detectors in order to prevent the illegal relocation and export of machines.

In response to this request, we, Yamazaki Mazak, have decided to install machine relocation detectors in our machines to prevent the machine from being illegally relocated to third country.

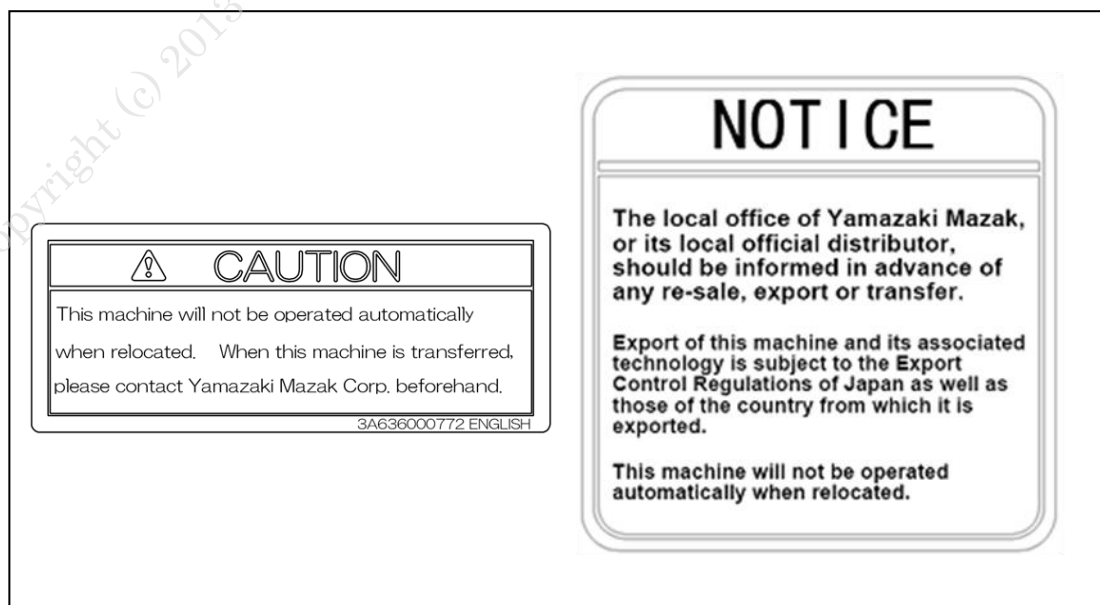
The current law enacted by the Japanese government in this control specifies that the reexport, reselling and relocation of specified machines are subject to prior approval of the Japanese government. If this is the case, please consult the Mazak Technical Center or Technology Center.

1-2 Procedure to Follow when Relocating the Machine

If you are planning to relocate your machine, contact the Mazak Technical Center or Technology Center without delay and let them know the scheduled date of relocation.

Note that relocating the machine without prior notice will make it incapable of operating automatically. Also, it might make it impossible to reuse the machine, because particular examinations are required to enable automatic operation.

Caution plate that indicates precautions to take when relocating the machine is attached in the vicinity of the main power breaker for the machine.



1

INTRODUCTION



Handwriting practice area with horizontal dashed lines.

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2 ALARM LIST

The machine relocation detector issues a total of 5 alarms, as listed below.

No.	Alarm message	Causes	Operation condition	Action
1393	R.D. Relocation Detected	Issued if the machine is replaced.	The machine cannot be operated automatically.	Contact Mazak Technical Center or Technology Center. At that time, you may be asked whether and how the machine was relocated.
1394	R.D. Communication Error	Issued if the machine relocation detector is out of order or its wiring is faulty.		
1395	RD Battery Change Time Exceeded	Issued if power supply to the NC unit is cut off for 30 minutes or more during replacement of the batteries.		
1397	R.D. Battery Not Mounted	Issued if the batteries are exhausted or the batteries are removed.		
1396	R.D. Low battery	Issued if the batteries are low.	The machine can be operated normally.	Purchase commercially available batteries and replace the batteries with new ones.

Note 1: One month after the alarm **1396 R.D. Low battery** is displayed on the NC unit, it becomes impossible for the machine to be automatically operated. This period is given as a guide, but not to guarantee a grace period of one month. So, if the alarm **1396 R.D. Low battery** is displayed, immediately replace the batteries with new ones.

Note 2: To prevent the battery from running out, leave the main power breaker ON as far as possible during the period between when the alarm **1396 R.D. Low battery** is displayed on the NC unit and when the batteries are replaced with new ones, even when the machine is not used.

Note 3: To change the batteries, follow the procedure in "3 HOW TO REPLACE THE BATTERIES." Before replacing the batteries, be sure to turn on the main power breaker and NC unit.



3 HOW TO REPLACE THE BATTERIES



- Handling the batteries with wet hands may result in an electric shock.

Note : Never remove the machine relocation detector when replacing the batteries. Otherwise, the machine will stop working and you will have to contact the Mazak Technical Center or Technology Center to restore it to normal operation.

3-1 Preparation

Before starting replacement of the batteries, have the following handy.

D size alkaline batteries (Recommended battery: D50FB00273*, Type: LR20RS(2S)JEE)

Note 1: Do not use manganese batteries.

Note 2: Always use two new D size alkaline batteries of the same type.

Note 3: Do not use rechargeable batteries.

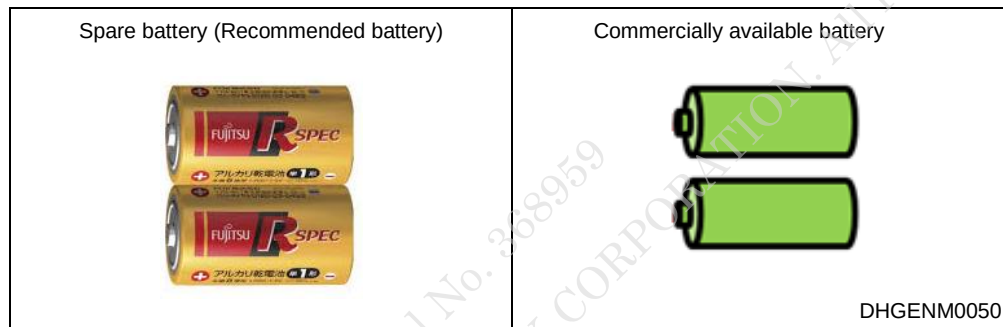


Fig. 3-1 Spare battery

3-2 Replacement Procedure

Note : Replace the batteries when the main power breaker is on, or when the NC power supply is on.

Name	Manufacturer's type	Yamazaki Mazak Parts No.
D size alkaline Battery	LR20RS(2S)JEE	D50FB00273*

- (1) Turn on the main power breaker or NC power.
- (2) Remove the cover from the battery box.



Fig. 3-2 Cover of the battery box

- (3) Replace all the two batteries with new ones.

Note : Install them in correct polarity as indicated below.



Fig. 3-3 Arrangement of batteries in battery box

- (4) Tighten the screws in a crisscross pattern to secure the batteries, while holding down the cover.

Note : If the main power breaker is turned off while the batteries are not installed correctly, the alarm **1395 RD Battery Change Time Exceeded** will be displayed after 30 minutes or so and it will become impossible to operate the machine automatically.

- (5) After changing the batteries, turn off the NC power (if it is on).
- (6) Turn off the main power breaker.
- (7) Turn back on the main power breaker.

- (8) Turn on the NC power.
- (9) After the NC unit has started, check to see that the alarm **1397 R.D. Battery Not Mounted** or **1396 R.D. Low battery** is not displayed.

If the alarm **1395 RD Battery Change Time Exceeded** does not go out even after the above operation is performed, contact Mazak Technical Center or Technology Center.

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